Here is the complete, updated textual database design based on all our final decisions:

- **Single PostgreSQL database** (no per-user SQLite files)
- All user data scoped by `user_id`
- Master tables are global (super_admin only)
- User-scoped tables have `user_id + admin_source` flag for sync/fork logic
- Organization branding included
- All extended fields from previous discussions (pins_json, notes, product_website, cable lengths, etc.)

1. Master Tables (Global Catalog – Super Admin Only)

```
SQL

-- MASTER TABLES (global, admin-controlled)

CREATE TABLE device_types (
    id SERIAL PRIMARY KEY,
    name TEXT NOT NULL UNIQUE,
    color TEXT NOT NULL DEFAULT '#3366FF',
    thumbnail TEXT,
    created_at TIMESTAMPTZ DEFAULT NOW(),
    updated_at TIMESTAMPTZ DEFAULT NOW()
);

CREATE TABLE manufacturers (
    id SERIAL PRIMARY KEY,
    name TEXT NOT NULL UNIQUE,
    website TEXT,
    created_at TIMESTAMPTZ DEFAULT NOW(),
    updated_at TIMESTAMPTZ DEFAULT NOW()
);

CREATE TABLE connection_types (
    id SERIAL PRIMARY KEY,
    name TEXT NOT NULL UNIQUE,
    color TEXT NOT NULL,
    spec_level TEXT,                    -- e.g., '2.1', '12G-SDI'
    created_at TIMESTAMPTZ DEFAULT NOW(),
```

```
    updated_at TIMESTAMPTZ DEFAULT NOW()
);

CREATE TABLE cable_lengths (
    id SERIAL PRIMARY KEY,
    length_value TEXT NOT NULL UNIQUE,  -- e.g., '10 m', 'Custom'
    display_order INTEGER DEFAULT 0,
    is_custom BOOLEAN DEFAULT FALSE,
    created_at TIMESTAMPTZ DEFAULT NOW(),
    updated_at TIMESTAMPTZ DEFAULT NOW()
);

CREATE TABLE master_node_types (
    id SERIAL PRIMARY KEY,
    device_type TEXT NOT NULL REFERENCES device_types(name),
    manufacturer TEXT NOT NULL REFERENCES manufacturers(name),
    model TEXT NOT NULL,
    pins_json JSONB NOT NULL,                  -- Extended pin format
    is_network_host BOOLEAN DEFAULT FALSE,
    is_network_switch BOOLEAN DEFAULT FALSE,
    notes TEXT,
    product_website TEXT,
    created_at TIMESTAMPTZ DEFAULT NOW(),
    updated_at TIMESTAMPTZ DEFAULT NOW()
);
```

2. User-Scoped Tables (Per User Data)

```
SQL

-- USER-SCOPED TABLES (filtered by user_id)

CREATE TABLE users (
    id SERIAL PRIMARY KEY,
    username TEXT UNIQUE NOT NULL,
    password_hash TEXT NOT NULL,
    role TEXT DEFAULT 'user',                  -- 'user' or 'super_admin'
    org_id INTEGER,
    last_synced_at TIMESTAMPTZ,
    created_at TIMESTAMPTZ DEFAULT NOW()
);

CREATE TABLE user_node_types (
```

```
    id SERIAL PRIMARY KEY,
    user_id INTEGER NOT NULL REFERENCES users(id) ON DELETE CASCADE,
    admin_source BOOLEAN DEFAULT FALSE,       -- TRUE = from master
    source_id INTEGER,                        -- master_node_types.id
    device_type TEXT NOT NULL,
    manufacturer TEXT NOT NULL,
    model TEXT NOT NULL,
    pins_json JSONB NOT NULL,
    is_network_host BOOLEAN DEFAULT FALSE,
    is_network_switch BOOLEAN DEFAULT FALSE,
    notes TEXT,
    product_website TEXT,
    created_at TIMESTAMPTZ DEFAULT NOW(),
    updated_at TIMESTAMPTZ DEFAULT NOW(),
    UNIQUE(user_id, device_type, manufacturer, model)
);

CREATE TABLE graphs (
    id SERIAL PRIMARY KEY,
    user_id INTEGER NOT NULL REFERENCES users(id) ON DELETE CASCADE,
    name TEXT NOT NULL,
    graph_json JSONB NOT NULL,                -- Full React Flow
serialization
    created_at TIMESTAMPTZ DEFAULT NOW(),
    updated_at TIMESTAMPTZ DEFAULT NOW(),
    UNIQUE(user_id, name)
);

CREATE TABLE systems (
    id SERIAL PRIMARY KEY,
    user_id INTEGER NOT NULL REFERENCES users(id) ON DELETE CASCADE,
    name TEXT NOT NULL,
    system_integrator_name TEXT,
    pins_json JSONB NOT NULL,                 -- Exposed system pins
    subgraph_json JSONB NOT NULL,             -- Internal graph
    is_network_host BOOLEAN DEFAULT FALSE,
    notes TEXT,
    created_at TIMESTAMPTZ DEFAULT NOW(),
    updated_at TIMESTAMPTZ DEFAULT NOW()
);
```

3. Organization Tables

```sql
-- ORGANIZATION TABLES

CREATE TABLE organizations (
    id SERIAL PRIMARY KEY,
    name TEXT NOT NULL UNIQUE,
    subscription_status TEXT DEFAULT 'active',
    created_at TIMESTAMPTZ DEFAULT NOW()
);

CREATE TABLE organization_branding (
    id SERIAL PRIMARY KEY,
    organization_id INTEGER NOT NULL UNIQUE REFERENCES organizations(id)
ON DELETE CASCADE,
    company_name TEXT NOT NULL,
    address TEXT,
    phone TEXT,
    email TEXT,
    website TEXT,
    tagline TEXT,
    logo_filename TEXT,
    logo_updated_at TIMESTAMPTZ,
    updated_at TIMESTAMPTZ DEFAULT NOW()
);
```

4. Key Indexes & Constraints

```sql
CREATE INDEX idx_user_node_types_user_id ON user_node_types(user_id);
CREATE INDEX idx_graphs_user_id ON graphs(user_id);
CREATE INDEX idx_systems_user_id ON systems(user_id);
CREATE INDEX idx_master_node_types_device_type ON
master_node_types(device_type);
```

5. Example Initial Data

```
SQL

-- Master data examples
INSERT INTO device_types (name, color, thumbnail) VALUES
    ('Switcher', '#3366FF', 'switcher.png');

INSERT INTO cable_lengths (length_value, display_order, is_custom) VALUES
    ('1 m', 10, FALSE), ('5 m', 50, FALSE), ('10 m', 60, FALSE),
('Custom', 999, TRUE);

INSERT INTO master_node_types (device_type, manufacturer, model,
pins_json, notes)
VALUES ('Switcher', 'BlackMagic', 'ATEM Mini Pro',
        '{"input_1": ["HDMI In 1", "HDMI", "2.0", "4K@60Hz"]}',
        'Located in Studio B');
```

Key Design Notes

- **Pin JSON format**: `{"input_1": ["name", "type", "spec_level", "connector_resolution"]}`
- **Sync logic**: Uses `admin_source` flag + `source_id` to safely merge master updates
- **Isolation**: All queries filter by `user_id = current_user.id`
- **Future**: Add `org_hidden_node_types` table for org-level hiding

This is the **final, consolidated PostgreSQL schema**.

Would you like me to:

- Generate the SQLAlchemy model classes next?
- Start Sprint 0 code (Flask + React skeleton)?
- Add more tables (e.g., user_branding, org_hidden_items)?

Let me know how you'd like to proceed!